When Does Boosting Beat Bagging?

From-Scratch Ensembles Under Noise, Imbalance, and Structural Variation

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***Abstract*—Boosting and bagging both combine weak or unstable learners, but they address different sources of error. We implement a weighted multiclass classification tree, discrete and real-valued AdaBoost, and a Random Forest from first principles, then evaluate their behavior through seven reproducible experiment families. A leakage-safe pipeline supports numeric and categorical data, severe imbalance, binary and multiclass targets, and deterministic execution. The supervised study is complemented by centered principal component analysis (PCA), K-Means, and DBSCAN, with bonus Gradient Boosting and t-SNE analyses. The completed full run covers Breast Cancer, Adult Income, and a 10,000-row Covertype subset. AdaBoost leads Random Forest on Breast Cancer five-fold accuracy (0.9666 versus 0.9561), while Random Forest leads on Adult (0.8649 versus 0.8556) and Covertype (0.7938 versus 0.6150). Under 20% training-label noise, the forest also loses less accuracy on Breast Cancer. These results support a conditional conclusion: boosting wins when sequential correction finds reliable residual structure, whereas bagging is preferable when model instability, noise, imbalance, or multiclass structure favors variance reduction. One command generated all raw fold, stage, corruption, bootstrap, and clustering evidence.**

I. Introduction

Boosting and bagging are often presented as interchangeable routes to "a stronger ensemble." Their mechanisms are not interchangeable. AdaBoost fits a sequence of weak learners to adaptively reweighted data; a sample that was classified incorrectly receives greater influence in the next round. Random Forest fits many high-variance trees to independent bootstrap samples and decorrelates them with node-level feature subsampling. Averaging then cancels idiosyncratic variation. The empirical question is therefore not which method is universally best, but under what data conditions each error control mechanism dominates.

This project asks: *under what conditions does boosting outperform bagging, and vice versa, and why?* We answer with controlled changes in ensemble size, tree depth, training-label noise, resampling, dimensionality, imbalance, and cluster structure. Unlike a black-box benchmark, the work is designed for oral defense: model state, class ordering, probability alignment, random seeds, out-of-bag (OOB) membership, and every bootstrap prediction are inspectable.

Our contributions are fourfold. First, we provide readable NumPy implementations of the assessed algorithms, including weighted split search. Second, a single configuration-driven runner produces all results and a machine-readable provenance manifest. Third, tests connect hand calculations and edge semantics to numerical reference trends. Fourth, optional extensions add SAMME.R, binary log-loss Gradient Boosting, t-SNE, and an executed interactive notebook without replacing any required experiment.

II. Algorithms and Mathematical Design

A. Weighted Classification Tree

For node class proportions p_k, Gini impurity and entropy are

$$I_G = 1 - \sum_{k=1}^{K} p_k^2, \qquad I_H = -\sum_{k:p_k>0} p_k \log_2 p_k. \tag{1}$$

With sample weights w_i, $p_k = \sum_{i:y_i=k} w_i / \sum_i w_i$. For a candidate split into children L and R, the implemented reduction is

$$\Delta I = I(P) - \frac{W_L}{W_P} I(L) - \frac{W_R}{W_P} I(R). \tag{2}$$

Each feature is stably sorted once at a node. Candidate thresholds are midpoints between successive distinct values. Incremental class-weight histograms make each threshold evaluation constant in K after sorting, rather than rebuilding both children. Equal gains resolve by feature order, then threshold order; leaf prediction resolves by sorted class order.

Root depth is zero. Hence `max_depth=0` produces a constant root and `max_depth=1` is a decision stump. Growth also stops for a pure node, too few samples, no positive-gain split, a degenerate child, or the depth limit. An XOR test intentionally confirms the standard greedy limitation: because each root split has zero gain, an ordinary CART tree does not invent a look-ahead split. Per-node feature sampling is independently repeated, and weighted impurity decrease supplies normalized feature importance.

B. AdaBoost: SAMME and SAMME.R

At round m, the stump's weighted error is

$$\epsilon_m = \sum_i w_i^{(m)} \mathbf{1}\{h_m(x_i) \neq y_i\}. \tag{3}$$

For K classes, discrete SAMME assigns

$$\alpha_m = \eta \left[\log \frac{1-\epsilon_m}{\epsilon_m} + \log(K-1) \right], \tag{4}$$

where η is the learning rate, then updates

$$w_i^{(m+1)} \propto w_i^{(m)} \exp\left(\alpha_m \mathbf{1}\{h_m(x_i) \neq y_i\}\right). \tag{5}$$

The final class maximizes $\sum_m \alpha_m \mathbf{1}\{h_m(x) = k\}$. Errors are clipped by a named numerical tolerance; a perfect learner stops early. A learner is rejected at error 0.5 for binary data and $1 - 1/K$ for multiclass data. The bonus SAMME.R option aggregates centered, clipped stump log-probabilities. All returned probabilities are score-derived and are not claimed to be calibrated.

Boosting attacks bias in an operational sense: subsequent stumps spend more capacity on unresolved regions. This can sharpen a weak boundary, but it can also allocate increasing influence to corrupted or intrinsically ambiguous observations. Staged prediction permits a single fitted maximum ensemble to generate the complete learning curve.

C. Random Forest

For each tree, n training indices are sampled with replacement. The complement is recorded as its OOB set. Every node considers a fresh random feature subset (default $\sqrt{p}$), so even trees trained on similar samples become less correlated. Class-probability columns from trees that omit a class are aligned into the forest's global class order before averaging.

If individual trees have prediction variance σ^2 and average pairwise correlation ρ, the variance of a mean of M trees is approximately

$$\rho\sigma^2 + \frac{1-\rho}{M}\sigma^2. \tag{6}$$

Bagging reduces the independent component; feature subsampling targets ρ. OOB accuracy uses only samples with at least one OOB voter and reports coverage separately, avoiding an implicit prediction for zero-voter rows. Child seeds are derived before work dispatch. Spawned Windows workers and the deterministic sequential fallback therefore produce identical trees.

D. Unsupervised Pipeline

PCA centers training data using μ, forms

$$\Sigma = \frac{1}{n-1}(X-\mu)^\top (X-\mu), \tag{7}$$

and orders covariance eigenvectors by descending eigenvalue. The explained variance ratio is $\lambda_j / \sum_\ell \lambda_\ell$; the smallest prefix reaching 90% is retained. K-Means minimizes

$$J = \sum_{i=1}^{n} \|x_i - \mu_{c_i}\|_2^2. \tag{8}$$

Ten deterministic k-means++ restarts are compared for each $k = 1, \ldots, 10$. An empty centroid is reseeded to the point with largest current squared error. DBSCAN defines $N_\varepsilon(x) = \{z : \|z - x\|_2 \leq \varepsilon\}$, including x itself. Core points have at least `min_samples` neighbors; reachable non-core points are borders; the remainder is noise. A provisional noise point may later become a border point during density expansion.

III. Data and Leakage Controls

Table I summarizes the locked sources. Breast Cancer is a binary medical baseline with 30 real features. Adult introduces categorical encoding, missing values, and moderate imbalance. A deterministic stratified Covertype subset preserves all seven classes and 54 features. The loader verifies the *realized* minority proportion; it enlarges the Covertype sample if the rarest class is not at most 1%, rather than assuming that a proposed sample meets the requirement.

TABLE I
Locked project datasets and intended stress factors.

Dataset	Task	Structural role
Breast Cancer	binary	569 × 30; balanced baseline
Adult Income	binary	48,842 × 14; mixed/missing
Covertype subset	7-class	10,000 × 54; minority 0.47%

All holdout and cross-validation protocols split first. Numeric medians, means, scales, categorical modes, and one-hot vocabularies are fitted only on the current training partition. Unknown Adult categories map to an all-zero block. Deterministic random oversampling is also confined to training data: a class below 5% of the original fold size is sampled with replacement up to that threshold, without reducing any majority class. PCA, clustering choices, noise corruption, and probability ordering never use test labels for model fitting or threshold selection.

IV. Experimental Protocol

The default seed is 42. Baseline models use a stratified 80/20 split and report accuracy, macro F1, and binary AUC or macro one-vs-rest multiclass AUC. The custom unpruned tree must fall within two absolute accuracy points of the sklearn tree reference. AdaBoost scaling spans estimator counts from 1 through 200 using staged predictions. Forest scaling evaluates counts through 200 at unrestricted depth and 100-tree forests at depths 1–20, recording test and OOB accuracy plus OOB coverage.

The head-to-head comparison fits one tree, 100-stump AdaBoost, 100-tree Random Forest, and a 100-tree sklearn reference on identical stratified five-fold splits. Label-noise experiments corrupt training labels only at 0, 5, 10, and 20%, with five replicates and nested corruption indices. Thus degradation within a replicate cannot be attributed to an unrelated set of flipped rows.

For bias–variance analysis, 100 shared bootstrap index arrays retrain both ensembles against one fixed stratified Breast Cancer test set. If $\hat{p}_b(x)$ is the probability vector from replicate b, $\bar{p}$ its mean, and e_y a one-hot target, we report

$$\mathrm{Bias}^2 = E_x \|\bar{p}(x) - e_y\|_2^2, \tag{9}$$
$$\mathrm{Var} = E_{x,b} \|\hat{p}_b(x) - \bar{p}(x)\|_2^2. \tag{10}$$

Their sum reconstructs expected multiclass Brier loss; the residual is stored as a numerical audit. The unsupervised study produces cumulative-variance, elbow, and k-distance curves; PCA scatter plots colored by true, K-Means, and DBSCAN labels; adjusted Rand index (ARI); and DBSCAN noise fraction. ARI is an external comparison, not an assumption that classes must form clusters.

TABLE II
FIVE-FOLD FULL-RUN ACCURACY (MEAN ± SD).

Dataset	Tree	AdaBoost	Custom RF	sklearn RF
Breast Cancer	.910±.021	**.967**±.020	.956±.015	.954±.016
Adult	.814±.002	.856±.003	**.865**±.003	.854±.004
Covertype	.714±.008	.615±.030	.794±.012	**.811**±.013

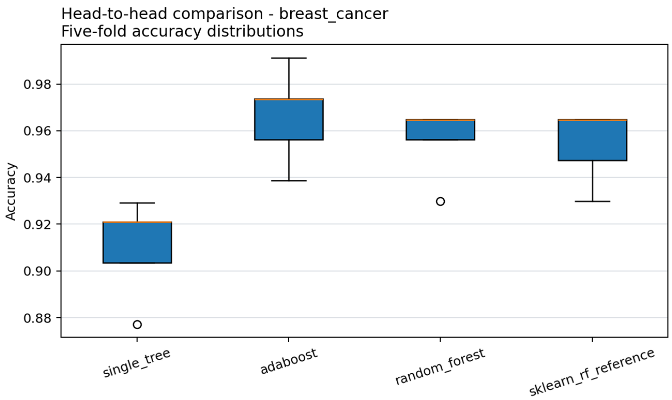

Fig. 1. Full five-fold Breast Cancer comparison; raw fold values are stored.

TABLE III
ACCURACY AT CLEAN AND 20% NOISY TRAINING LABELS (MEAN ± SD).

Dataset	AdaBoost		Random Forest	
	Clean	20%	Clean	20%
Breast Cancer	.956±.000	.896±.039	.954±.004	.939±.009
Adult	.857±.000	.855±.002	.869±.001	.861±.002
Covertype	.593±.000	.663±.006	.787±.003	.781±.003

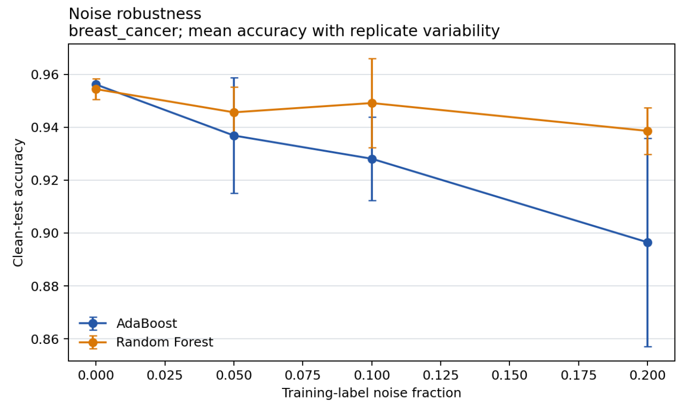

Fig. 2. Full five-replicate noise experiment with 100-estimator ensembles.

V. Full-Run Results and Interpretation

The deterministic no-argument run completed all 28 phases for Breast Cancer, Adult, and Covertype. The manifest records `profile=full`, seed 42, the exact data versions, and the complete grids. The runner generated 82 artifacts in 6.88 hours; all values below come from those stored CSV, JSON, and NPZ files.

A. Correctness, Scaling, and Head-to-Head Behavior

On the 80/20 baseline, custom/reference tree accuracy was 0.9123/0.9123 on Breast Cancer, 0.8110/0.8102 on Adult, and 0.7155/0.7235 on Covertype. Every absolute gap is below the required two percentage points, including the largest gap of 0.0080 on Covertype.

Table II answers the research question conditionally. AdaBoost is best on the smaller continuous Breast Cancer task, also leading macro F1 (0.9639) and AUC (0.9934). The custom forest leads the assessed models on Adult with macro F1 0.7995 and AUC 0.9175. On severely imbalanced seven-class Covertype, the custom forest substantially exceeds AdaBoost in accuracy (0.7938 versus 0.6150), macro F1 (0.6769 versus 0.3742), and AUC (0.9623 versus 0.8484). The sklearn forest is retained only as reference and is slightly stronger there.

Scaling curves reinforce this distinction. Breast Cancer AdaBoost reaches its minimum 0.0351 test error by 10 rounds and retains it at 200. Adult AdaBoost reaches its minimum 0.1406 test error at 175 rounds. Covertype instead reaches its best 0.3530 error at 10 rounds and deteriorates to 0.4405 at 200, consistent with later reweighting being unhelpful. Forest accuracy stabilizes by roughly 50–100 trees: the 200-tree results are 0.9561, 0.8685, and 0.7900 on the three datasets. Depth optima are shallow for Breast Cancer (3) but deep for Adult and Covertype (19).

B. Noise Robustness and Bias–Variance

On Breast Cancer, AdaBoost loses 5.96 accuracy points by 20% noise, compared with 1.58 for the forest; its replicate variability is also larger. Adult is less sensitive for both models, while the forest remains ahead at every noise level. Covertype is non-monotonic: corrupted-label AdaBoost improves over its clean fit but remains far below the forest. This unexpected result is reported rather than converted into a general robustness claim; multiclass reweighting, oversampling, and a weak clean ensemble can make random perturbation act like regularization.

The 100 shared bootstrap fits in Table IV separate probability behavior from classification accuracy. Random Forest has much lower Brier bias squared, whereas AdaBoost has lower variance but conservative, uncalibrated vote probabilities. Residual magnitudes below 4×10^{-17} verify the stored decomposition.

C. Unsupervised Structure and Bonuses

PCA retains 7, 21, and 36 components to reach 90% variance. Breast Cancer has moderate K-Means alignment, but Adult and Covertype ARIs near 0.10 show that class labels do not coincide with centroid geometry. DBSCAN is weakly aligned on every dataset under the deterministic k-distance policy. This is valid negative evidence: supervised classes are not required to be density clusters.

SAMME.R improves accuracy on Breast Cancer (0.9561 to 0.9649) and Adult (0.8573 to 0.8687). On Covertype, accuracy rises from 0.5930 to 0.6700 but macro F1 falls from 0.3458

TABLE IV
BRIER DECOMPOSITION OVER 100 SHARED BOOTSTRAP FITS.

Model	Bias^2	Variance	Expected loss
AdaBoost	.3382	.0007	.3388
Random Forest	.0675	.0046	.0721

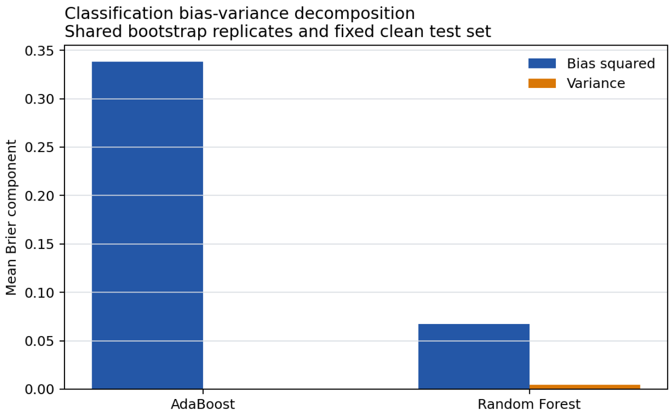

Fig. 3. Full probability/Brier decomposition on a fixed clean test set.

to 0.2946 and AUC from 0.8302 to 0.7870, demonstrating why accuracy alone is unsafe under imbalance. Gradient Boosting reaches AUC 0.9899 on Breast Cancer but lower accuracy than AdaBoost (0.9386 versus 0.9561); on Adult it underperforms on both accuracy and AUC. t-SNE remains a local visualization only; global distances and apparent cluster areas are not used as evidence.

VI. Reproducibility, Tests, and Limitations

The command `python src/experiments/run_all.py` completed the full suite in 6.88 hours. The manifest stores the seed, complete configuration, package versions, dataset sources and distributions, local hashes, output paths, and Git commit when available. The original runtime recorded 28 completed phases. A final smoke test later overwrote top-level timing metadata and the Breast Cancer output family; the test was isolated, the full Breast Cancer artifacts were regenerated, and the incident is recorded in `validation_report.md`. Numeric metrics were unchanged and remain deterministic.

Fifty-one tests cover hand-computed splits and AdaBoost weights, arbitrary labels, repeated values, zero sample weights, depth semantics, progress cleanup, numerical clipping, OOB complements, class-column alignment, voting ties, sequential and parallel identity, PCA centering and orthogonality, K-Means empty clusters, DBSCAN boundary and noise reassignment, fold-local preprocessing, oversampling, result schemas, and end-to-end artifact creation. The measured suite covers 88.71% of source statements; every core algorithm exceeds 85%. Ruff and mypy complete without findings.

Limitations are explicit. Exhaustive split search is educational rather than optimized for industrial scale. The fixed-test-set bias–variance decomposition has finite-sample uncertainty and estimates probability loss, not a universal classification decomposition. Random oversampling can duplicate rare observations and does not synthesize within-class variation. Automatic DBSCAN ε selection is deterministic but cannot eliminate domain judgment. Finally, OOB and cross-validation results depend on the selected source version and stratified subset; the manifest is essential context.

TABLE V
FULL UNSUPERVISED ANALYSIS ON DETERMINISTIC SAMPLES.

Dataset	PCs	k	KM ARI	ε	DB ARI	Noise
Breast Cancer	7	3	.515	3.483	.030	6.50%
Adult	21	4	.100	2.218	.092	5.10%
Covertype	36	4	.095	3.463	.067	1.85%

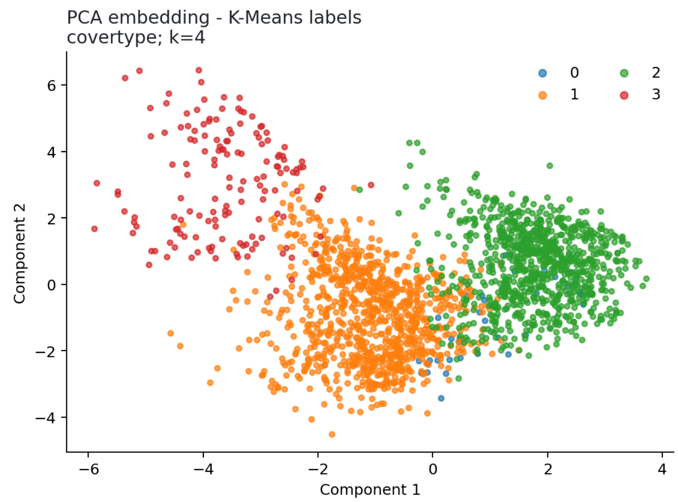

Fig. 4. Covertype PCA view colored by $k = 4$ K-Means assignments; geometry is descriptive and does not claim recovery of seven classes.

VII. Conclusion

The full results reject a universal winner. Boosting is strongest on Breast Cancer, where reliable residual error allows sequential stumps to improve the boundary. Bagging wins on Adult and decisively on multiclass, severely imbalanced Covertype; it is also more robust to Breast Cancer label noise and produces much lower Brier bias. The Covertype noise anomaly and SAMME.R's accuracy/macro-F1 disagreement show why mechanism, class distribution, and metric choice must be interpreted together. Unsupervised ARIs further confirm that label structure need not match centroid or density geometry.

The project supports these claims with auditable implementations rather than library wrappers. Its most important engineering result is the connection between theory and evidence: weights, OOB memberships, stage predictions, bootstrap probabilities, and clustering labels are all retained so that every conclusion can be defended and reproduced.

AI Assistance Disclosure

AI tools assisted code scaffolding, tests, documentation, and artifact generation. The human team remains responsible for mathematical verification, full-result review, truthful contributions, and defense. No library ensemble is concealed inside an assessed implementation.

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